Reflection Report

Project: Student Information Management System (SIMS) **Student:** [Student name]

> Honesty note: the entries below are real events from the build session. Add your own observations from your oral defence preparation where marked.

1. AI tools used

Claude Code with Claude Sonnet 5.5 (agentic CLI). Headless Chrome was used only to capture screenshots.

2. Prompt strategy

Start with one broad instruction so the agent reads all requirement files, then steer with short, single-purpose prompts (one UI concern each). When a prompt was ambiguous ("no borders on x, only y") the agent stated its interpretation instead of guessing silently. Audits ("is it still in the instruction?") were used as validation prompts.

3. Agent strategy

One agent, several sequential roles (planner, database, Laravel, UI, testing, documentation). See `agent-design.pdf`. Limitation: roles were not independent reviewers, so the human review step matters more.

4. What worked

- Reading the rubric first and building toward its weights (CRUD, API, auth, tests, docs).
- Running the test suite after every change; 32 tests pass.
- Turning repeated UI into Blade components, which made later UI requests one-file changes.
- Verifying with evidence (route list, rendered pages, restore test) rather than trusting "it should work".

5. What failed

- Large shell scripts creating many files at once failed with `unexpected EOF`; the work was re-done in smaller batches and with the file-writing tool.
- A Python patch script mangled Windows-style backslash namespaces (`\N` escape error); replaced by a direct file write.

- First screenshot attempt wrote files to a wrong path because of backslash handling; fixed with forward slashes.
- The folder rename failed because a `php artisan serve` process held the folder.

6. AI mistakes -> correction -> validation (D2 evidence)

AI suggested	Problem found	Correction applied	Result validated
Migrations generated with identical timestamps (`enrollments` sorted before `subjects`)	Foreign key to a table that did not exist yet	Renamed migration files so `subjects` runs before `enrollments`	`migrate:fresh --seed` succeeds
`role` in the User `$fillable` (typical scaffold)	Registration form could set `role=admin` (privilege escalation)	Removed `role` from `$fillable`; set only in seeder; DB default `staff`	`test_user_can_register_as_staff` sends `role=admin` and asserts `staff`
`unique` validation rule reused for update	Editing a record without changing its email would fail as "already taken"	`Rule::unique()->ignore($id)` in the shared Form Request	`test_update_allows_keeping_own_unique_values`
Test used `$data + ['status' => 'completed']`	PHP `+` keeps the left array's keys, so status stayed `enrolled`; test failed	Switched to `array_merge`	Full suite green
Default Laravel pagination view	Unstyled (Tailwind markup without Tailwind)	Custom pagination view + styles; later limited to 5 numbers	Rendered pages 1,2,3,5,8,9 of a 10-page set
User said "80rem max-width"	The CSS value was actually 1100px	Reported the difference, removed the cap as requested	Tests pass; layout fills width

AI suggested	Problem found	Correction applied	Result validated
Search on subjects using `where(...)->orWhere(...)`	`orWhere` outside a group would break any additional filter	Wrapped in a nested closure	Search + paginate pages render

7. Human corrections

The human directed every UI change, chose the interpretation of ambiguous requests, decided how git should be handled for the final owner (init only, no fabricated commit history), and stopped the server process that blocked the rename. **[Add your own corrections here.]**

8. Lessons learned

1. AI output must be verified with commands and tests, not by reading it.
2. Small, single-purpose prompts are cheaper, safer and easier to review than one giant request.
3. Security defaults (mass assignment, authorisation on every destructive route) are easy for AI to miss; ask for them explicitly and test them.
4. State ambiguity back to the requester instead of silently choosing.
5. Honest documentation matters: this repository does not claim an AI comparison, deployment or commit history that did not happen.
6. **[Add what you would improve with two more weeks: e.g. enrollment capacity limits, grade reports, Postman collection, deployment.]**